EXPERIMENT 1: INSTALL AND CONFIGURE AI LIBRARIES

Aim

To configure the Python environment in Google Colab and verify the availability of NumPy, Pandas, Transformers, and PyTorch.

Algorithm

1. Start Google Colab.
2. Import the required libraries.
3. Display the installed library versions.
4. Check whether GPU acceleration is available.
5. Display the configuration results.

Program

Step 1: Install the required libraries.

Run this cell in Google Colab.

%pip install -q transformers

Google Colab generally provides NumPy, Pandas, and PyTorch already. We install Transformers if needed.

Step 2: Execute the following program.

```
# Experiment 1: Configure AI Libraries

import sys

import numpy as np

import pandas as pd

import torch
```

```
import transformers

# Display Python version

print("Python Version:", sys.version.split()[0])

# Display library versions

print("NumPy Version:", np.__version__)

print("Pandas Version:", pd.__version__)

print("PyTorch Version:", torch.__version__)

print("Transformers Version:", transformers.__version__)

# Check GPU availability

if torch.cuda.is_available():

    print("GPU Available: Yes")

    print("GPU Name:", torch.cuda.get_device_name(0))

else:

    print("GPU Available: No")

    print("Using CPU")

print("All required libraries are working.")
```

Input

No external input is required. The program checks the installed Python environment and libraries.

Expected Output

Python Version: 3.x.x

NumPy Version: 2.x.x

Pandas Version: 2.x.x

PyTorch Version: 2.x.x

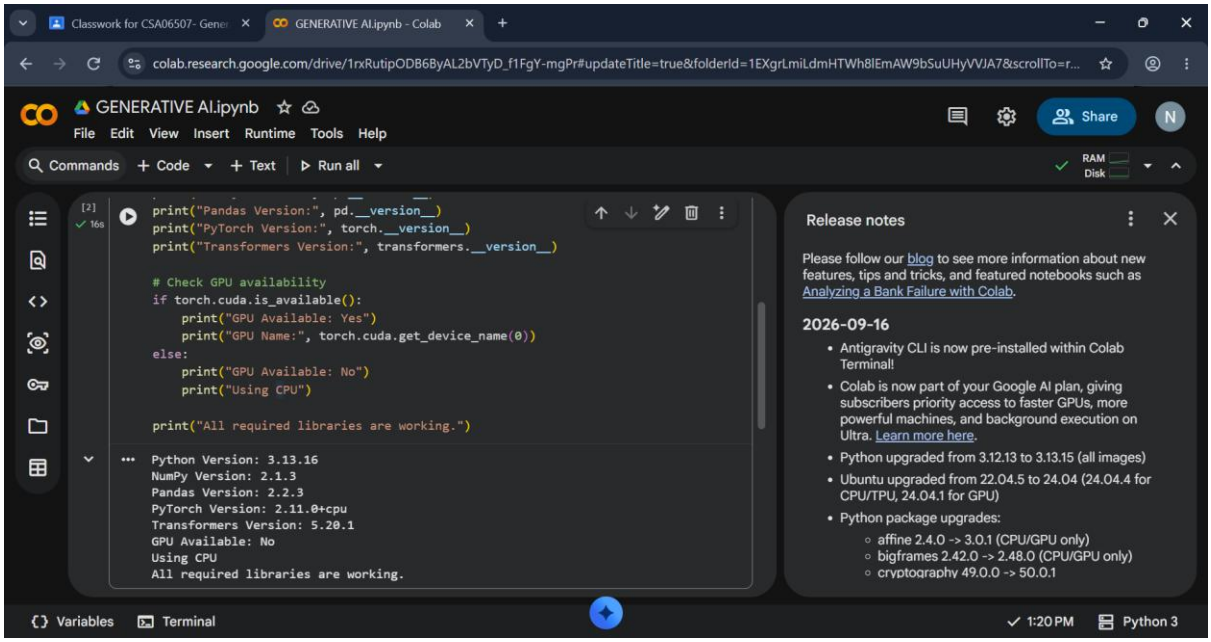
Transformers Version: 5.x.x

GPU Available: No

Using CPU

All required libraries are working.

The versions shown are illustrative. Your output will display the actual installed versions. GPU availability depends on your Colab runtime.



Result

Thus, the Python environment was configured in Google Colab, and the availability of NumPy, Pandas, Transformers, and PyTorch was verified successfully.